Chaeun Lee

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RESEARCH INTERESTS

World Models · Generative Models · Embodied AI · Efficient AI Systems

I want to understand how an AI system can turn experience into a theory of the world. Rather than memorizing tasks or predicting the next state alone, I am interested in world models that form generalizable insights from experience, uncover the mechanisms behind them, and recombine those insights to solve unfamiliar tasks and environments. I am particularly interested in how such models can update their understanding through physical interaction and support embodied agents that must reason and act under uncertainty. I also want to study how model architectures and their execution can be designed together so that increasingly capable reasoning models remain efficient under real compute, memory, and data-movement constraints.

EDUCATION

Sungkyunkwan University (SKKU) **Mar 2023 - Feb 2027**

B.S. in Systems Management Engineering | Second Major: Advanced Semiconductor Engineering | Expected Feb 2027

- Overall GPA: 4.02 / 4.5 | Major GPA: 4.10 / 4.5
- Academic Excellence Scholarship: Fall 2024, Spring 2025
- Relevant Coursework: Parallel Computer Architecture & Programming; AI System Semiconductor Design; AI System Design

PUBLICATION

S. Choi, W. Choi, S. Chun, D. Lee, J. Kim, **C. Lee**, H. Woo. "Efficient Skill Grounding via Code Refactoring with Small Language Models." *International Conference on Machine Learning (ICML), PMLR 306, 2026.* [arXiv:2606.07999]

SELECTED RESEARCH & ENGINEERING EXPERIENCE

π_0/VLA Inference Optimization on BOS NPU **Sep 2026 - Present**

BOS NPU / SKKU

- Ported and optimized π_0 Vision-Language-Action inference on a BOS NPU using the TTNN / TT-Metalium software stack, covering both the VLM backbone and flow-matching action expert.
- Traced the end-to-end π_0 inference path at the layer and operator level, analyzing intermediate tensor shapes, activation dependencies, and K/V state generation and reuse across VLM processing and iterative flow-matching action refinement.
- Investigated heterogeneous VLA workloads and dataflow bottlenecks, examining how tensor shapes, compute mapping, memory placement, and repeated execution patterns affect performance across the VLM and action expert.
- Developing and evaluating dataflow-aware mapping and execution optimizations, including Static-Dynamic KV Attention and operator fusion, to reduce redundant computation and memory transfers during on-device VLA inference.

Machine Learning and Mind Lab (MLML), KAIST **Jun 2026 - Aug 2026**

Research Intern | Advisor: Prof. Sungjin Ahn

- Reproduced the Generative Recursive Model (GRAM) on N-Queens, Graph Coloring, and Sudoku, implementing data generation, training, and evaluation pipelines.
- Decomposed reasoning performance into solution generation and selection, examining implementation details in KL-loss computation, dataset construction, and training/evaluation procedures.
- Investigated how stochastic latent noise affects reasoning accuracy and solution coverage through controlled experiments, including inference-time noise scaling.
- Explored alternative reasoning and search mechanisms using concepts from SMC, flow-based generation, and fixed-point reasoning.

CSI-Agent Lab, Sungkyunkwan University **Oct 2025 - May 2026**

Undergraduate Research Intern | Advisor: Prof. Honguk Woo

- Implemented the RepairAgent baseline for embodied code-as-policy research, adapting its generate-repair-revise workflow to robot skill transfer across different embodiments and environments.
- Built and evaluated agent execution and repair pipelines in RLBench and Genesis, investigating cases where generated code executed without errors but failed to achieve the intended manipulation task.
- Analyzed failure cases under partial observability, where the observed failure point did not necessarily reveal the underlying cause and prior robot actions had already altered the scene, sometimes leaving partial task progress intact.
- Structured current scene information, object/gripper states, execution history, task constraints, and failure signals as feedback for local repair, continuation from post-failure states, and revision of remaining behavior while preserving completed task progress.
- Evaluated the baseline across robot transfer settings, analyzing task success, subtask failures, repeated repair attempts, token usage, and execution latency.

EARLIER RESEARCH EXPERIENCE

System Intelligence Lab (SILAB), KAIST **Jun 2025 - Aug 2025**

Undergraduate Research Intern | Advisor: Prof. Jinkyoo Park

- Studied deep reinforcement learning, model-based RL, world models, diffusion models, and flow-based generative models through research seminars, paper discussions, and independent study, including DreamerV3 and V-JEPA.
- Conducted Diffusion Posterior Sampling (DPS) experiments for denoising and inpainting inverse problems and presented the results in a lab seminar.

SELECTED RESEARCH & TECHNICAL PROJECTS

EmbodiedInsightBenchmark **Feb 2026 - Mar 2026**

Collaborative Research Project (2 months) | BEHAVIOR-1K / OmniGibson

- Designed and implemented simulation-based tasks evaluating two of six embodied reasoning dimensions: Context Adaptability (CA) and Scenario Flexibility (SF).
- Context Adaptability (CA): Developed tool-use and manipulation scenarios requiring agents to recognize relevant similarities and differences between previous experiences and current physical conditions, then adapt their actions accordingly. Examples include selecting a spatula to retrieve an egg from a narrow gap and adjusting scraper orientation for grill cleaning.
- Scenario Flexibility (SF): Implemented a rule-based scenario involving a power outage during cake preparation, modeling changes in appliance temperatures, power restoration, and time-dependent task constraints. Developed symbolic state transitions and evaluation logic to compare alternative action plans and their possible outcomes.

AI-Based Bad Pixel Detection & Correction

STOB League 2025 | Team Project

- Developed a 3-head U-Net for joint bad-pixel detection, correction, and Quad-Bayer demosaicing in CMOS image sensors.
- Built training and evaluation pipelines for clustered bad-pixel correction and joint demosaicing.
- Applied INT4 GPTQ quantization and SparseGPT pruning for NPU-oriented deployment.
- Achieved 11.43× model compression, 87.5% lower memory usage, and 63.6% lower inference latency, with approximately 1 dB PSNR degradation.

CUDA Kernel Optimization

Parallel Computer Architecture & Programming

- Implemented tiled GEMM using 32×32 shared-memory tiles, achieving up to 1.96× speedup over a global-memory baseline through increased data reuse.

AI Accelerator RTL Implementation

AI System Semiconductor Design

- Converted a PyTorch MNIST DNN into 16-bit fixed-point SystemVerilog RTL, including memory placement and inter-layer data movement.

AWARDS & LEADERSHIP

Grand Prize (President's Award), SKKU Convergence Foundation Project

The Knowledge Center for Innovative Higher Education, SKKU | Team Leader | Aug 2024 | KRW 2.4M Award

- Led the team from problem formulation and idea development through prototyping, final presentation, and Q&A.
- Developed and demonstrated an eco-friendly labeling concept based on the whitening behavior of plastic materials.

OTHER ACTIVITIES

Amateur Radio Club (HAM), SKKU

Executive Committee Member

- Planned and organized semesterly fox-hunting (radio direction-finding) activities, including event design and radio-equipment selection.

TECHNICAL SKILLS

Programming: Python, C++, CUDA, SystemVerilog
Machine Learning: PyTorch, generative models, recursive reasoning, model compression, quantization
Embodied AI / Simulation: RLBench, Genesis, BEHAVIOR-1K / OmniGibson
AI Systems: TTNN, TT-Metalium, NPU profiling, CUDA kernel optimization, memory/layout optimization